CHAPTER 11

Party Line

Sputnik was a press agent's dream come true. But it was apotheosis to a totalitarian state boasting of technological progressivism in a decade when the exploits of jet aircraft circling the world nonstop or polar icebreakers and submarines daring the Arctic titillated the public. The Russian people, too, must have felt justifiable pride. Despite Stalinist claims that everything from the telephone to baseball had been invented by Russians, the West tended to denigrate their achievements even in fields, like nuclear physics, where they had contributed.[1] Now the Soviets were leaders in an awe-inspiring technology that was arguably a Russian preserve since the days of Tsiolkovsky. It would have been peculiar had the Soviet regime *not* celebrated its feat in hyperbolic terms, especially since Sputnik seemed strikingly relevant to every major plank of the 1956 Party platform.

True, Khrushchev may have been no more aware than Eisenhower of the impact the first satellite would have. But once evident, it must have seemed so ideal a conjuncture as to validate the premier's reading of the material laws at work in the history of his time. In a matter of days and weeks after Sputnik every propaganda theme of the next seven years was coined and in circulation. First, sputniks verified the Soviet claim to an ICBM, hence the new vulnerability of the United States to nuclear assaults. Second, the space achievement irrefutably demonstrated the scientific, and ultimately economic, superiority of the USSR. Third, the Soviet space program was dedicated solely to peaceful conquest of the cosmos in the interest of all mankind. Fourth, the leading inspiration and guide of the Soviet space effort was none other than Premier Khrushchev himself, and it illustrated his dynamic leadership.

"When we announced the successful testing of an intercontinental rocket," Khrushchev told James Reston a week after *Sputnik I*, "some American statesmen did not believe us. The Soviet Union, they claimed, was saying it had something it did not really have. Now that we have successfully launched an earth satellite, only technically ignorant people can doubt this. . . ." The Soviet ICBM was "fully perfected," and the USSR already possessed "all the rockets it needs of all ranges."[2] The

United States, on the other hand, clearly did not have intercontinental rockets or it would have launched its own sputnik. Instead, the United States was planning only a tiny satellite weighing eleven kilograms. *Sputnik II* weighed 508 kilograms and "if necessary, we can double the weight of the satellite."[3] The military implications were made clear by the end of November: "The fact that the Soviet Union was the first to launch an artificial earth satellite, which within a month was followed by another, says a lot. If necessary, tomorrow we can launch ten, twenty satellites. All that is required for this is to replace the warhead of an intercontinental ballistic rocket with the necessary instruments. There is a satellite for you."[4]

During the months when Eisenhower was trying to reassure Americans, Soviet disinformation sought to persuade them that the balance of power had suddenly and irreversibly shifted. *International Affairs,* a Soviet journal published in English, proclaimed the U.S. bomber force obsolete. What was more, U.S. overseas bases were now subject to Soviet rocket attacks and served only to involve other countries in the destruction sure to follow U.S. adventures.[5] Khrushchev repeatedly stressed the risks attached to continued alliance with the United States and spoke of states being "wiped from the face of the earth" in the event of war.[6]

The second theme—Sputnik as proof of the superiority of socialism—emerged in the very first press release from *Tass,* and was elegantly expanded by *International Affairs* the following spring. History, it proclaimed, knew few examples of a scientific discovery giving its name to a whole epoch, but this was surely one: "We can say that the era of the conquest of the Cosmos has dawned." But Sputnik was a "many-sided phenomenon." Could so amazing an advance in scientific and technical thought be separated from its social roots? Could it arise independent of industrial, scientific, and technical development in the sputniks' mother country? And if the sputniks were indeed a social phenomenon, did this not mark the dawn of a new era in international relations? *International Affairs* answered its own questions: yes, the sputniks proved the social maturity of the USSR, and yes, such a breakthrough did revolutionize the social and political development of other peoples. Just as the steam engine had signaled the maturity of capitalism, the sputniks were the mark of an emerging socialist system. As the French Communist physicist Frédéric Joliot-Curie observed, "It is no accident that the Soviet Union was the first to launch a sputnik. In this fact lies the law of development of the new society. This outstripping of Western science will each year become more frequent." Hence, the most telling fact about Sputnik was its parentage: "scientists, engineers, and workers reared by our Communist Party in accordance with an advanced ideology. The whole world saw yet one more extraordinary, important demonstration of the Socialist system's superiority to the capitalist system."[7]

Of course, the West would eventually have satellites, too, granted

International Affairs. American militarists were already scheming to place weapons in space and on the moon. "Marxist science long ago established the truth . . . that only in a socialist society do scientific discoveries and technical achievements serve human progress and the welfare of mankind." Socialist priority in space was a potent force for peace. Furthermore, the sputniks reaffirmed the correctness of Marxist materialism by refuting "idealistic theories that the world is 'unknowable,' that the objective world does not exist but is only a product of the human consciousness. . . . The appearance in outer space of man-made satellites also proved the stupidity of attempts to combine science with clericalism, with religious theories regarding the divine origin of the world." The sputniks also enhanced the moral and political influence of the USSR and have "taken the ground from under the feet of malicious propaganda regarding the alleged economic and cultural backwardness of our country. It showed the world the Soviet Union as it really is."[8]

Finally, there were the military implications of the sputniks. All American strategies of terror—"situations of strength," Cold War, Truman Doctrine, containment, liberation of "captive peoples," brinksmanship, massive retaliation—all these varieties of "atomic blackmail" were bankrupt. Imagine the threat to the world, the author shuddered, if the United States had retained its atomic monopoly and become sole possessor of ICBMs as well! But the innate superiority of socialism ensured that that nightmare did not come to pass. Instead, "The world has entered a new stage of co-existence . . . *when any attempt by the imperialists to launch a new world war will inevitably boomerang against the entire capitalist system and lead to its complete downfall.*"[9]

The Communist Party line depicted the space program, unencumbered by any artificial division between civilian and military programs, as entirely peaceful. Despite Khrushchev's talk of the interchangeability of H-bombs and sputniks and the utter secrecy surrounding the Soviet program, Marxist science was deemed progressive and in the interest of all. Candid U.S. statements about the military uses of satellites, on the other hand, were carefully gathered, polished, and flung back by the Soviet agitprop slingshot. Khrushchev accused the United States of seeking to militarize the cosmos, warned of devastating Soviet ripostes, and demanded the dismantling of all (now "useless") foreign bases. American refusal to comply then permitted Khrushchev to ignore American requests for technical study of orbital disarmament.

Throughout the period of the missile gap (1958 to 1961) Khrushchev punctuated his ultimata against the Western presence in Berlin with "rocket rattling," made credible by the space shots. On November 10, 1958, Khrushchev unilaterally renounced the Potsdam agreement on four-power control in Berlin and threatened to recognize East Germany's claims to sovereignty. Four days later he boasted that Soviet ICBMs were rolling off the assembly line.[10] The Luniks were "irrefutable proof" that the USSR could land a hydrogen bomb on a kopeck anywhere in the world. *Luna III,* launched on the second anniversary of *Sputnik I,* signaled Khrushchev's arrival in the United States in October 1959. Nevertheless, the peaceful nature of all Soviet space missions continued to be stressed. *Korabl Sputnik I,* according to *Pravda,* was "a peaceful explorer of the secrets of nature. . . . in the brilliance and glory of our science, the light of its noble aims, the bandit flights of the American spy plane [U-2] look pitiful and lowly before the whole world."[11] *Korabl Sputnik II* "delighted the whole world," but the "bourgeois scribblers" wasted no time concocting evil fabrications about Soviet intentions. "It is not on Soviet soil that one should look for madmen outlining plans for military bases in space. They are entrenched on the other side of the ocean, in the madhouse of the Pentagon, which has become the main asylum of spies and provocateurs."[12]

The ICBM was another matter. The Soviets did all they could in 1959 and 1960 to feed American panic over the "missile gap." Marshal Malinovsky thanked Khrushchev at the Twenty-first Party Congress for "equipp[ing] the armed forces with a whole series of military ballistic missiles [including] intercontinental." Marshal Grechko announced that the Red Army had "received" the ICBM, and Malinovsky praised its "pinpoint accuracy."[13] The "serial production" of ICBMs, coming off the line "like sausages," Khrushchev told the Party Congress, meant that "Socialism has triumphed not only fully, but irreversibly."[14] In May 1959 a delegation of West German editors was told that the USSR had enough bombs and rockets "to wipe from the face of the earth all of our probable opponents."[15]

The single most important audience for Soviet chest-thumping was Western Europe. The threats against foreign bases and allies of the United States cut directly at the bonds holding together the NATO alliance. Even American determination to resist, to go to the brink over Berlin, could be made to seem "insane" in light of Soviet prowess. While if the United States could be made to relent, and make even minor concessions, NATO might begin to unravel. Of course, Eisenhower's confidence in the continued value of U.S. bombers, the solidarity and nerve of his, Adenauer's, de Gaulle's, and Macmillan's governments, frustrated Khrushchev in Berlin.[16] But there is no question that the space campaign hurt American prestige. In June 1955 only 6 percent of West Europeans thought the West weaker than the USSR in military might. In November 1957, after two sputniks, 21 percent in Britain thought the West weaker, 20 percent in France, 12 in Italy, and 10 in Germany. But the United States, taken alone against the USSR, was deemed inferior by 50 percent of the British and a quarter of the French.[17] By 1960, after two years of missile propaganda, the Soviets were judged superior in military strength to the United States by 59 to 15 percent in Britain, 37 to 16 in France, 45 to 15 percent in Norway, and 47 to 22 percent even in loyal West Germany (the remainders judged the powers equal or were undecided).[18] More troubling still was European opinion of global trends. An April 1960 poll revealed that a plurality of Europeans in every country expected the USSR to be stronger than the United States after twenty years of "competition without war." Only a fourth of the British and one Frenchman in fourteen thought the United States could prevail over that long haul![19] Soviet propaganda was so demoralizing that the impression continued to prevail in Britain (and even in the United States) that the Soviets had launched more (as well as bigger) satellites than the Americans, long after it ceased to be true.

Secrecy, the most potent weapon of the totalitarian state, permitted the deception and fed Western imaginations. The origins of the Soviet rocket program, its chief designers, the uses to which they planned to put their heavy sputniks, imminent launch plans, even the location of their test sites, design bureaus, and missile plants were hidden. Launches were not announced beforehand, failures not at all. Official spokesmen predicted space stations and lunar flights, but provided no details or schedules, or estimates of Soviet spending. Soviet scientists at international conferences were often third-raters and presented papers based on published material or data from foreign sources. The configuration of the great booster, designated by NATO the SS-6 "Sapwood," was itself unknown for some time. All this fed the anxious expectations of Westerners, waiting and wondering what startling new capability would be revealed by the Soviets' next "surprise."

The burden of concocting the next surprise fell to the unacclaimed hero of Soviet engineering, Chief Designer Korolev. He was awarded additional bureaus, factories, and laboratories, which he fashioned into a national space complex. But he won no honors or fame, and had to divert his talent toward "spectaculars" for which Khrushchev took credit and used to back the party line and rocket-rattling. But the "ol' number seven" rocket had demonstrated its maximum payload. The addition of an upper stage permitted the moonshots of 1958 and 1959. The next spectacular, by which the relative standings in space would surely be measured, was launching a man into orbit. Korolev was expected to hurry no less than NASA's Mercury executives.

The lead time necessary to any venture involving new hardware, especially life-support systems, suggests that research on manned spaceflight began even before Sputnik. High-altitude flights of animals dated from the earlier 1950s, *Sputnik II* carried the dog Laika, and in 1958 an obsolete aircraft factory near Moscow was turned over to Korolev and converted into a "manned spaceflight center." Certainly Project Mercury, underway by the end of 1958, lit a fire under the Soviet man-in-space program in more ways than one. According to defecting journalist Vladimirov, a special translation bureau existed to gather and distribute all data from U.S. sources on spaceflight. For two years Soviet staffs pored over American articles on manned rocket flight and life support.

The Vostok program, though built on indigenous foundations, thus grew in parallel with Mercury. Cosmonaut recruitment began in 1959, as did design and fabrication of a man-rated upper stage for the SS-6 and the spherical space capsule itself. Given the rudimentary state of Soviet space facilities in these years, the devolution of responsibility for all space missions on the same teams, and the political pressure to schedule launches around anniversaries and political events, one can imagine the pressure building on Korolev and his lieutenants. Khrushchev complicated matters by insisting, apparently for reasons of security, that capsules be brought down on Soviet ground instead of in the sea as the Americans intended. This required a far heavier and sturdier craft, not to mention a prodigious parachute system, all adding considerable weight. The makeshift solution was to arrange for the pilot to eject after reentry and descend on his own parachute, allowing the capsule to land at a much higher velocity.[20]

The first test of this Korabl Sputnik system was in May 1960; the second was when Strelka and Belka were recovered in August. Predictions appeared in the Western press of a Soviet manned flight. But behind the veneer of boastful self-confidence, the harried Soviets charged with maintaining appearances suffered several crippling blows that put the United States back in the chase. And if the Americans should succeed in getting the first man into space, much of Sputnik's impact might be undone. It was a very near thing.

On September 19, 1960, the Soviet liner *Baltika* docked in New York harbor. On board was Premier Khrushchev and in his luggage were some "miniature spaceships," presumably to be trotted out during his stay in celebration of some space triumph. A moon shot heralded his visit the previous year; this journey coincided with a "launch window" for Mars. Three rockets were apparently prepared, but nothing happened. Khrushchev even extended his time in New York, grew more bellicose with passing days, and finally pounded his shoe on the table at the UN. He left for home on October 13, and several days later, Soviet tracking ships were observed to turn homeward as well. Still nothing happened.

Over the years accounts of launch failures and even a catastrophe trickled to the West. Intelligence leaks, Soviet defectors, the double-agent Oleg Penkovsky, and Soviet historian Zhores Medvedev all testified to the worst space-related disaster in history, attributable in part, according to Medvedev, to Khrushchev's "misuse of space research to boost Soviet political prestige." Apparently, two Mars-bound rockets fizzled after launch on October 10 and 14. By the evening of the twenty-third, with the "window" about to close, the third rocket failed to ignite. Field Marshal Mitrovan I. Nedelin, commander of the new Strategic Rocket Forces, ordered technicians to leave their blockhouses and examine the rocket close-up. Suddenly the propellants in the great rocket exploded.

MERCURY-REDSTONE 2 AND VOSTOK I

In mid-January 1961 both contestants in the race for manned spaceflight seemed to be having their troubles. The Soviets aimed directly at orbital flight, but their Korabl Sputniks had had uneven success. The U.S. Mercury program aimed first at the more modest goal of "shooting a man from a cannon" up into space and back into the sea 300 miles downrange. Still, an American first man in "space," even if not in orbit, might steal the acclaim from a subsequent cosmonaut's orbital flight. It was well within the realm of possibility. The smaller, more thoroughly tested Redstone could be man-rated far more quickly than the giant, pressurized Atlas booster that would serve for later orbital missions. The army's Jupiter had carried two monkeys, Able and Baker, on a suborbital trajectory as early as May 1959. But the first Mercury/Redstone mating (November 1960) had flopped.

On December 19, the *MR (Mercury/Redstone)-1A* performed well, reaching an altitude of 131 miles. *MR-2* went up on January 31, 1961, and inside it was another "astronaut"—a chimpanzee named Ham. Several malfunctions resulted in the chimp spending four hours on the ground strapped inside the Mercury, blasting up to a velocity a thousand feet per second more than expected, enduring a load of 17 g's and a cabin pressure that dropped, thanks to a faulty valve, from 5.5 to only one pound per square inch, six and a half minutes of weightlessness, and a landing 130 miles beyond the target, and thus having to bob in the Atlantic for two hours forty minutes before retrieval by a helicopter. Free at last and aboard ship, Ham sampled an apple and an orange, and generally behaved like a chimp. The engineers were unhappy with the eccentric behavior of *MR-2.* With the sturdiness of primate physiology they could only be pleased.

The tentative schedule pointed to launch of *MR-3,* with a man aboard, in late March. Quick fixes had been made at Huntsville in response to *MR-2,* and the seven Mercury astronauts were eager to go. But safety was at least as important to NASA's political and budgetary position as rapid results, and the seven planned modifications suggested another test flight. On February 13 at Huntsville, the decision was made to keep *MR-3* and astronaut Alan Shepard waiting until late April at the earliest. The extra test, *MR-BD,* went off without a hitch on March 24 and got the MR system "man-rated." But it could have carried a man.

Korolev and the Vostok team must have applauded the doughty Ham. How much they learned from the data made available by the open NASA program is debatable. But a healthy chimp, combined with two Korabl Sputnik recoveries in March, must have boosted Korolev's confidence considerably. On April 12, in the midst of the period between the extra Mercury test and Alan Shepard's moment, Korolev lit the engines of his modified ICBM and flung Yuri Alekseyevich Gagarin into orbit about the earth. He circled the globe in ninety minutes, completed his assigned tasks, and was brought back into the atmosphere by ground control, landing from his parachute in a pasture in central Russia. The Vostok capsule—minus the cosmonaut—came to rest in a "plowed field" near the Leninsky Put collective farm not far from the towns of Marx and Engels on the banks of the Volga. A very appropriate spot.

Many (estimates range from scores to hundreds) of the Soviet Union's most skilled space personnel perished in a thunderous fireball on the pad in Kazakhstan. Korolev and Yangel, now his deputy, were indoors and survived. Marshal Nedelin, who gave his name to the disaster, did not. Two days afterward Moscow reported that Nedelin had died—in a plane crash. Only in 1979 did an official biography substitute for this a vague account that he had died "tragically in the performance of his official duties." Khrushchev's memoirs admit that Nedelin was killed in a missile "malfunction."[21]

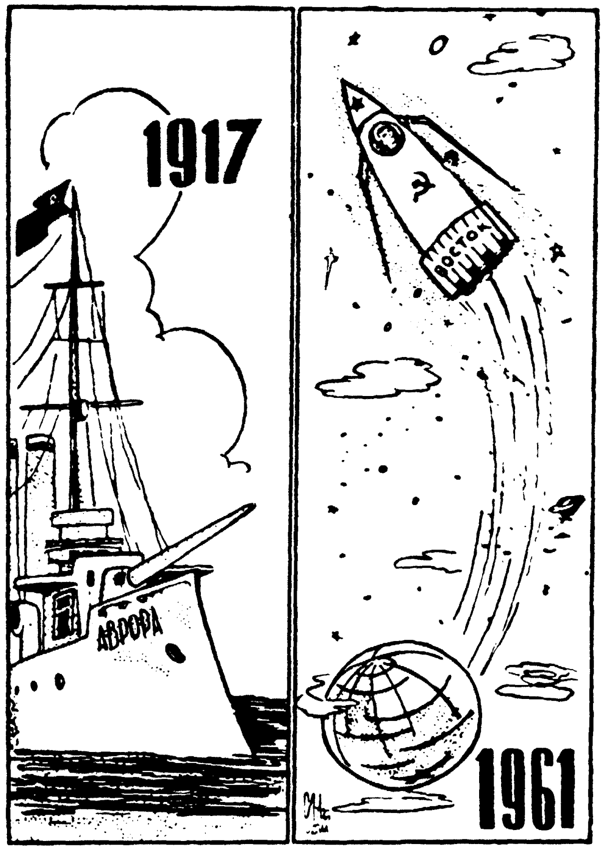
The loss of men and machines, though extensive, could be made up. The effects on morale, on Korolev's nerves and conscience, and on his relationship with Khrushchev were probably more damaging and irreversible. But the pressure for continued space triumphs only increased: NASA was now projecting suborbital manned flights by the spring of 1961. Korolev wanted to fly tests with primates, like the Americans, but was not given the time. Then on December 2, 1960, *Korabl Sputnik III,* with dogs Pcholka and Mushka, reentered the atmosphere at too sharp an angle and the capsule burned up. The next day Korolev had a heart attack. Doctors also diagnosed a kidney disorder, a common ailment in survivors of the *gulag,* and prescribed a lengthy rest.[22]

Whether on his own drive, the knowledge that continued funding hinged on beating the Americans, or on callous orders from Moscow, Korolev went back to work. Two more test flights failed in February. Redesign and preparation of two more proceeded at a breakneck pace. *Korabl Sputniks IV* and *V* were recovered successfully in March. If his luck held, Korolev would soon give Khrushchev a great gift, another four or five years of life to the myth of Soviet space leadership, and the party line based on it.

Yuri Gagarin's flight was a second Sputnik (or third if one counts the 1959 Lunik that helped jolt Ike into competing for prestige). Three and a half years of American protestations and accomplishments could not stand before this latest "proof" of Soviet technological might. When the first American blasted off three weeks later, his suborbital trajectory and "sloppy splashdown" seemed a poor imitation of the Soviet feat. The space gap, in the eyes of the world, had widened.

Vostok I was a measure of the genius of Korolev, the competence of Soviet engineers, and the courage of Gagarin. But it was said to "testify to the scientific, social, economic, and moral superiority of the Socialist system," while "increas[ing] immeasurably the strength and authority of the Socialist world in its struggle to insure a peaceful future for the inhabitants of our planet."[23] To Khrushchev, Gagarin's flight marked

> a stage of the scientific/technical development of our country and reveals its mighty upward flights from backwardness to progress. In this flight are reflected the heroic accomplishments of the Soviet people, the working class, collective



"The Dawn (Aurora) Always Heralds a New Day." The cartoon plays on the names of the naval ship *Aurora* (the dawn), a cradle of Russian Revolutionary agitation, and *Vostok* (the East), the first manned spacecraft, implying that the future is always made in the Soviet Union. From *The Morning of the Cosmic Era* (Moscow, 1961).

farm peasantry, working intelligentsia, and our wonderful scientists, who have created a technical present-day miracle, the cosmic ship Vostok, this greatest triumph of the immortal Lenin's ideas.[24]

The very name "Vostok," a word like "Orient" that meant "upward flow" and may have been chosen by Korolev to signify the "upward flow" of humanity, was translated for foreigners as "The East," to suggest the rising sun of communism.[25]

To *International Affairs,* the real secret of Soviet spaceflight was

> rooted in the specific features of Socialist society, in its social structure, its planned economy, the abolition of exploitation of man by man, the absence of racial discrimination, in free labor and the released creative energies of peoples. Our achievements in the field of technology in general and in rocketry in particular are only a result of the Socialist nature of Soviet society.[26]

Gagarin himself, it was said, sent greetings during his hasty trip over

Africa to the peoples below struggling to break the chains of imperialism.

Success and world acclaim could not induce the Soviet leadership to lower its curtain of secrecy. Instead, they set a pattern of official lies and coverups about the space program that lasts to this day. Korolev remained a nonperson, guarded by security agents and given his medals in secret lest, it was said, he be abducted or assassinated by the CIA. Academicians Keldysh and Sedov, and especially Khrushchev himself, were paraded as the masterminds of the space program. They even fudged the details of *Vostok I,* claiming that Gagarin had ridden his craft to the ground, a stipulation of international flight records. The mysterious Soviet space launch complex was—like all place names on Soviet maps—given an erroneous location. The great cosmodrome was said to lie near the town of Baikonur. In fact, U-2 flights and ingenious private spacewatchers in Japan and England, who backtracked sputnik trajectories, established the location some 200 miles away near the railhead of Tyuratam, whereupon this bustling Kazakh "Huntsville" and "Canaveral" all in one disappeared from Soviet maps and gazeteers.[27]

Propaganda, like any advertising, appeals to unconscious fears and desires. For every Western observer who concluded that Soviet secrecy hid falsity and backwardness, many more thought the Soviet space program all the more sinister. According to the State Department, *Vostok I* earned "extraordinarily heavy" media coverage in Europe, perhaps greater than *Sputnik I.* While the note of astonishment was no longer present—the Soviets merely confirming their lead in space—the overall impact was as great. Even conservative journalists in Europe granted Soviet leadership and speculated uneasily on the use the Kremlin might make of its leverage. A poll taken after Shepard's flight in May revealed that West Europeans believed the USSR ahead in total military strength by 41 to 19 percent, and in overall scientific achievement by 39 to 31 percent.[28]

Reactions in Latin America, Africa, South Asia, and the Middle East varied with the political slant of nations and journals, but none disputed the Soviet claim to scientific leadership. An independent paper in Manila saw the real importance of Soviet cosmonautics in "its effect on the people of the uncommitted countries, who see in all this the supposed superiority of the Communist way of life, economic system, and materialistic philosophy." A conservative paper in Kuala Lumpur: "It is evident that the U.S. is losing the space race with Russia. . . ." Sukarno of Indonesia was "confident that the Soviet feat would eventually contribute to the progress and prosperity of mankind as well as to world peace." Nehru of India thought that it revealed the narrowminded folly of preparing for war and considered Vostok a victory for peace. The official journal of Iran thought Gagarin's flight more important than the discovery of the new world; that of Tunisia, more important than the invention of the printing press; that of Kenya, more important than the discovery of

— С наступающим днем, Африка!

"In Tune with the Times . . . Africa!" The cartoon depicts Yuri Gagarin saluting the African people from space, implying that each is engaged in the same, mutually supporting struggle against imperialism. From *The Morning of the Cosmic Era* (Moscow, 1961).

the wheel. Egypt's Nasser praised the "gigantic scientific capabilities" of the Soviet people and had "no doubt that the launching of man into space will turn upside down not only many scientific views, but also many political and military trends."[29]

The global impact of Soviet space shots was, of course, intangible. No uncommitted nation can be shown to have "chosen" socialism or allegiance with Moscow on the basis of the standings in the space race. But Soviet successes must have made the socialist model far more respectable than it was ten years earlier, when U.S. hegemony was assumed, and provided Third World intelligentsias with excuses and encouragement to lean toward a national socialism, neutralism, or anti-Americanism to which they tended for other reasons.[30] More interesting, perhaps, was the reverse phenomenon: the effect of Cold War rivalry in the Third World on the subsequent history of spaceflight. For just as Sputnik sparked great changes in U.S. domestic policy, so the narrow Soviet victory in inaugurating manned spaceflight helped to complete the Space Age revolution in the United States and, in the process, send men to the moon.

Space propaganda was equally important in Soviet domestic politics. The party line held that space triumphs proved that socialist science and technology must inevitably advance more rapidly, and had in fact overtaken the West. But the proof of the pudding in Communist theory was overall industrial production and technical sophistication. After all, it was the *imperialists* who tended to place new technology at the service of war, while socialist inventions were released to benefit the people. Hence the space shots supposedly symbolized the broad advance of technology throughout the Soviet economy. This was the case, according to Khrushchev, who boasted that the USSR would surpass the United States in per-capita production in the near future. But Soviet rocketry in fact reflected isolated, force-fed development in a few sectors of the militarized economy. In other words, space travel *was* the triumph of a command economy, but its fruits were bitter, not sweet. For, as Soviet nontraditionalists had always contended, broad economic progress depended on massive shifts of resources away from the military sectors to agriculture and light manufacturing. Thus, even as Khrushchev pointed to space technology as proof that communism would soon provide "all the needs of the people," the space and missile expenditures may have inhibited that very achievement.

The space propaganda was also designed to impress on foreign opinion the new and awesome military power of the USSR. Here was a second contradiction. For despite its "firsts," the USSR was in fact behind the United States in every meaningful category of missile technology. Even the size of the Soviet rocket was a measure of backwardness, for it made a poor ICBM. Third, Khrushchev perceived the Space Age as the age of the attainment of communism. As such, it meant a profound change in domestic Soviet organization. But insofar as these reforms relied on the expertise of the technical intelligentsia, they accentuated the old friction between technicians and the Party and between interest groups and Khrushchev himself. Desirous of consolidating his power and promoting his policies, Khrushchev set up the space program, much as Stalin had aviation, as a personal trophy. Hence he depended on the technicians, Korolev most of all, to provide him with glory and leverage, even as he frustrated them and twisted their efforts to serve his personal rule. From the Red Terror to Stalin's attack on "technocrats" and the aviation designers, to Khrushchev's clumsy interference with the space program, such was the pattern of Soviet-style technocracy.

Khrushchev claimed to be the inspiration and leader of the space program. He played "father" to the cosmonauts after the fashion of Stalin. And yet he showed little interest in the scientific benefits of spaceflight and apparently never attended a launch. But he obliged Korolev to set aside followup missions and move on at once to new "breakthroughs" more likely to stun the world. Hence the fitful quality of the early space programs in the USSR. The first Sputniks soon gave way to the Luniks, then to the manned program. If Korolev and his associates had begun to work on the various applications of space that the Americans undertook from the start—and later programs suggest that they had—they must have done so in their spare time. Korolev had to bow to Khrushchev's wishes or lose his budget and perhaps his position to envious rivals like Yangel or Chalomei. His health was broken, many of his comrades dead, and his resources and technology inferior to those available to his American competitors. How long could he maintain the fiction of Soviet leadership, and at what cost to the real progress of Soviet space technology? He must have yearned to attend foreign conferences, accept the acclaim due him, and consult with foreign colleagues about techniques for the conquest of space. But he could not. By the summer of 1961, he must also have sketched out a rational plan for future development. If so, he could not implement it. In mid-July Khrushchev summoned Korolev to his Black Sea dacha and informed him that the next manned flight must surpass the achievements of Gagarin, and it must take place within a month—to punctuate, as it turned out, the erection of the Berlin Wall on August 13.[31]

In his memoirs Khrushchev confessed, "Of course, we tried to derive the maximum political advantage from the fact that we were first to launch our rockets into space. We wanted to exert pressure on the American militarists—and also influence the minds of more reasonable politicians—so that the United States would start treating us better."[32] But the quest for "maximum political advantage" led to the espousal of a "party line" that not only hindered rapid and rational development of space technology but encouraged a dangerous deception in military policy as well.

CHAPTER 12

The Missile Bluff

For it was all a bluff. At the very time Khrushchev boasted of the obsolescence of American "massive retaliation," the U.S. deterrent was at the height of its effectiveness and the USSR had yet to deploy a single ICBM. It is not known whether the design of the first Soviet ICBM was approved just before or just after the first Soviet H-bomb test of August 1953.[1] Possibly the conventional wisdom to the effect that the ICBM was so big because it was built to accommodate atomic bombs is an oversimplification. The first Soviet H-bomb was itself a cumbersome affair, since it contained a much larger detonator of fissionable material than the U.S. bomb and its yield was estimated to be only half a megaton. Thus the Soviets may have assumed that mightier H-bombs would still require a large rocket.[2] The critical determinant of ICBM size, however, may instead have been the relative inefficiency of Soviet rocketry. Apparently unable to construct a jet nozzle capable of withstanding the high temperatures generated by a larger engine, Soviet engineers made do with clusters of the smaller RD-107 engine. This meant a lower thrust-to-weight ratio than in the first U.S. ICBMs and a shorter range. For all its size, the SS-6 had a range of only about 4,340 miles.[3] Thus giganticism in Soviet rocketry was a sign of primitivism, and the very clusters of clusters system that made the Soviets appear to be ahead in the 1950s inhibited their efforts to keep up in the 1960s.

What was worse, the SS-6 relied on an unstorable kerosene and LOX combination and could not be launched quickly. Nor could the unwieldy rocket be easily fitted in silos. Finally, as the Soviets had not mastered inertial guidance, the SS-6 had to be radio-guided at intervals in its boost phase. All told it was a decade behind the solid-fueled Minuteman already under design. Yet world opinion, including American, was quick to tab Sputnik a sign of Soviet superiority! What was the Kremlin to do? Throw a wet blanket on the otherwise delightful chorus of panic and praise by admitting that the USSR was years away from an operational missile? Or encourage the impression of Soviet might? Deploy the SS-6 despite its faults? Or wait wisely for a better product and finesse the widening gap between talk and truth?